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Book Title: Science and Technology in the Global Cold War

Book Editor(s): Naomi Oreskes and John Krige

Published by: The MIT Press

Stable URL: <https://www.jstor.org/stable/j.ctt9qf6k8.17>

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13 Big Science and “Big Science Studies” in the United States and the Soviet Union during the Cold War

Elena Aronova

In the 1960s, Big Science was identified as a new phenomenon, associated with changes in the organization of scientific research in the aftermath of World War II.¹ As the Big Science mode of research blossomed and expanded in the second half of the twentieth century, it became a widespread mode of scientific research in the natural sciences. More recently, in the social sciences, research has been increasingly organized around big projects involving big budgets and big interdisciplinary teams. Big Science has also become a favorite analytical category for historians of science who have used it for talking not only about Cold War science but also about its predecessors in earlier decades or even centuries. Indeed, the precedents of Big Science, understood as a mode of research characterized by large scale and extensive state involvement in its funding, could be found in the imperial sciences in the nineteenth century and in the great Victorian projects of knowledge that similarly had at their center the presence of big state as well as big business. As an intervention into this discussion, the historian Jon Agar has suggested distinguishing between the *phenomenon* of Big Science as a mode of organization of scientific research and the “*labeling* of ‘Big Science’ as something of concern, ... a product of the ... long 1960s.”² Endorsing this useful distinction, this chapter aims to historicize Big Science as an analytical category, showing how the categories we typically use to characterize science during the Cold War were themselves products of Cold War science.

In the early 1960s, the phenomenon of Big Science became a focus of debates among social and natural scientists, political analysts, and scholars of diverse disciplinary and political backgrounds. On both sides of the “Iron Curtain,” social analysts articulated awareness that the large-scale growth of science after World War II had significant implications, for better or for worse, for modern societies, identifying Big Science as a contemporary phenomenon requiring characterization and careful study, especially in terms of its social and political implications.

In the first section of this chapter I focus on the discussions in the United States concerning the political and social consequences of Big Science, which found its most prominent articulation in the early 1960s within the network of intellectuals

associated with Congress for Cultural Freedom. In the second section I focus on the parallel discussions in the Soviet Union, where the discussion of the same phenomenon was articulated in terms of the theory of Scientific Technological Revolution (STR), promulgated in the 1960s and the 1970s. As I discuss in the third section, these discussions provided a context for reading Kuhn's *Structure of Scientific Revolutions* (1962), both in the United States and in the Soviet Union, in those decades.

I argue in this chapter that Big Science, as an analytical category, was deployed as a resource to debate, negotiate, and rationalize the concerns and anxieties of the Cold War on both sides of the Iron Curtain. Throughout the Cold War, both the United States and the Soviet Union advocated their ability to offer and display different visions of modern industrial (or rather "post-industrial") society, and Big Science played major role in these powerful imageries. In their different political settings, American and Soviet scholars conceptualized the phenomenon of Big Science, tying together their country's preferred model of social order with an image of the "right" scientific order and, at the same time, creating parallel imageries of societies built around modern technology. In the context of the Cold War and the expansion of the sciences into Big Science, these scientists and scholars also helped to invent a set of new subjects while reconceptualizing science as a social activity and in various ways exploring science-society-politics nexus. In both political settings, the discussion of social and political implications of Big Science contributed to the construction of the public space in which studies of science (or, rather, of Big Science) emerged as an important—and politically relevant—area of expertise.

The "*Minerva* debate": Big Science and "Big Science Studies" in Cold War America

In an editorial in the first issue of *Minerva*, the journal's founding editor, the University of Chicago sociologist Edward Shils, described the scope and the aims of the new journal:

The governmentalisation of science and scholarship is, in part, a product of intellectual development and its changed relationship to technology, which entails costs which can only be borne by government, and returns, in which governments, even in capitalistic societies, have a great and appropriate interest. The governmentalisation of science in the past decade and a half is also, in part, a result of the Cold War—as are also, to some extent, the political embarrassments and concerns of science. *Minerva* will be concerned with the indirect as well as the direct influence of the Cold War on the role of science and scholarship and on the performance of their true calling.³

The "governmentalisation" of science, as Shils emphasized, implied newly defined relations between American scientists and the federal government. In the aftermath of World War II, in Shils' words, scientists had "become politicians."⁴ Politics was

always part of science, but now, Shils argued, the relationship between politics and science had become more explicit and prominent. This intensified interdependency between science and politics was manifested on different levels, ranging from the development of scientific advisory representation in the government to "scientists' movements" such as the Pugwash Conferences. The aim of *Minerva*, Shils stated, was to create a forum for discussing, describing, documenting, and examining these recent changes and how they affected the relations between science and politics: "Without seeking to formulate a code to govern them, [*Minerva*] will try, through the description of what has happened and through the analysis of the conditions of good and poor relationships, to develop a better understanding of why things sometimes go wrong and to contribute to a better practice."⁵

During the 1960s, *Minerva* developed into a forum for discussions of the changing relations between science, politics, and national policy in the United States. The first and the most prominent topic of the newly founded journal, which became a continuing theme in the first decade of its publication, was the phenomenon of Big Science and its social and political implications. In 1961, the term "Big Science" was simultaneously coined by two physicists—Alvin Weinberg, director of the Oak Ridge National Laboratory and a member of the President's Science Advisory Committee, and Derek de Solla Price, a physicist turned historian.⁶ In his influential 1961 essay in *Science* on the phenomenon that he called Big Science, Weinberg argued that the large-scale centralized scientific operations exemplified by gigantic National Laboratories such as Oak Ridge—which were linked to big facilities, big budgets, and big publicity—had dramatically changed the major features and societal role of science. While believing in the positive effect of Big Science and its advantages for American science, Weinberg argued that the funding for Big Science projects should be confined to National Laboratories, "to prevent the contagion" of Big Science from spreading to the universities. Big Science, in Weinberg's view, should be "segregated" from the more traditional modes of research in order to "prevent it from taking over Little Science."⁷

Weinberg's ambivalent position with regard to Big Science was widely shared by scientists of different disciplinary backgrounds who argued that large-scale, lavishly funded, team-operated research in Big Science threatened individual scientists' initiative and innovation.⁸ Moreover, Big Science, with its huge expenditures, had placed new and unprecedentedly large demands on society. As Weinberg noted, however, the right question to ask was not whether Big Science was "ruining science." Since Big Science was "here to stay," the more appropriate thing to do, in Weinberg's view, was to systematically examine and characterize this phenomenon and its social and political implications. Arguing that "Big Science was an inevitable stage in the development of science," Weinberg warned scientists and the public about the broad social and long-term consequences of Big Science as a contemporary cultural phenomenon, comparing it to the construction of the pyramids of Egypt or the Palace of Versailles.

Like those cultural landmarks, Big Science was, in Weinberg's view, "a supreme outward expression of our culture's aspirations" that had "created many difficult problems, both philosophic and practical."⁹ "These questions" addressing the cultural, social, and political implications of Big Science, Weinberg noted, "are so broad, and so difficult, that I cannot do more than raise them here."¹⁰

The questions that Weinberg sketched in his short 1961 *Science* essay were at the center of Shils' concerns when he founded *Minerva*. While planning the first issue of the journal, in May of 1962, Shils invited Weinberg to contribute an expanded version of his *Science* essay.¹¹ Weinberg apparently didn't respond, but he did respond to Shils' second written request several months later. With his letter Weinberg sent the text of the lecture he presented at the meeting of the local honorary engineering society at the University of Tennessee earlier that year.¹² Shils responded enthusiastically and suggested that it be published in *Minerva*.¹³

Weinberg's article—titled, at Shils' suggestion, "Criteria for Scientific Choice"—focused on the question of how funding for Big Science projects, requiring large federal patronage, should be allocated in a democratic society: Should scientists play a leading role in political decisions concerning their science, or should their role be limited to providing technical expertise to politicians?¹⁴ As a director of an exemplary Big Science operation closely working with national-security and military agencies, Weinberg didn't question the power that scientists, especially physicists in the wake of the World War II, had at influencing decision-making and obtaining large state resources while staying largely free from political control. The problem for Weinberg was that all decisions pertaining to Big Science were inevitably political, based more on judgments of politicians and few elite scientists, mostly physicists, than on diversified scientific or technical expertise. How, Weinberg asked, should the government decide what to prioritize when it came to "different, often incommensurable, fields of science"? "We shall," he wrote, "have to choose between, for example, high-energy physics and oceanography or between molecular biology and science of metals ... whose only common characteristic is that they all derive support from the government."¹⁵ The problem of such choices, Weinberg maintained, "arises in 'Big Science', not in 'Little Science.' ... It is only when science really does make serious demands on the resources of our society—when it becomes 'Big Science'—that the question of choice really arises."¹⁶ Weinberg's solution was to "decentralize" these choices by resorting to the "systematic application of a set of criteria." Weinberg suggested three groups of criteria: "technological merit" (the balance between the costs of research and the prospective return in the form of industrial applications), "scientific merit" (measured as much by indirect repercussions as by direct promise), and "social merit" (promise of high payoffs for health, food production, defense or prestige). Molecular biology, for example, in Weinberg opinion, had all three merits—"scientific," "technological," and "social"; physics was currently overrated; space research was only "masquerading" as

science: "If we do space-research because of prestige, then we should ask whether we get more prestige from a man on the moon than from successful control of the water logging problem in Pakistan's Indus Valley Basin. If we do space-research because of its military implications, we ought to say so. ..." ¹⁷

Weinberg's article, although mostly concerned with the pragmatic issue of the allocation of funds for Big Science projects, simultaneously framed a set of more general themes. Big Science, Weinberg emphasized, was explicitly political. In Weinberg's view, the qualitative change in the relation between science and the state had occurred when science achieved the size and complexity of Big Science operations, all of which were embroiled in institutional, bureaucratic, and national, as well as international, politics. Moreover, as Weinberg's discussion implicitly suggested, it was the Cold War that had set up this new agenda for science, with its incentive to maintain the nation's superiority in all fields that might contribute, directly or indirectly, to the high-technology postwar economy. Weinberg concluded that scientists and the general public had no choice other than to accept the Big Science, but that they should ask questions about its political and social consequences and try to come up with reasonable answers.

Weinberg's essay opened the discussion of the social and political implications of Big Science on the pages of *Minerva*. Shils carefully planned the debate. Along with the paper from Weinberg, he asked Michael Polanyi, whom Shils deeply admired, to contribute a paper for the opening issue. ¹⁸ Polanyi was an enthusiastic supporter of Shils' journal, praising Shils for the "great achievement" of his "enterprise in bringing out the first issue of *Minerva*" and assuring his friend that he "subscribed to everything you say in the editorial introduction." ¹⁹ The essay Polanyi wrote for the opening issue of *Minerva* was his famous "The Republic of Science." ²⁰

In his essay, Polanyi articulated his long-held views on science and its governance, arguing that the social order of science constituted a coherent and self-governing system, with no central authority, internally coordinated "by mutual adjustment of independent initiatives." This system, Polanyi held, "works according to economic principles similar to those by which the production of material goods is regulated." ²¹ As Polanyi put it in his famous science-as-market analogy, "in the free cooperation of independent scientists we shall find a highly simplified model of a free society." ²² According to this view, the governmental support of science, though welcome, should not in any way imply the control of scientific enterprise, either in the form of planning or regulation, since any form of state intervention would undermine the order of science based on the self-governing autonomy of the scientific community. Science, Polanyi argued, should maintain its traditional "Little Science" mode of organization, as a decentralized network of independent self-coordinated initiatives, since only this mode of organization could provide science with its crucial strength. ²³

The long-cherished ideal of the self-governing autonomy of science was, however, in conflict with the Big Science mode of organization of science, as Weinberg's discussion made clear. With its centralization of research and decision making through centralization of facilities Big Science obviously didn't conform to the image of science as a spontaneous pursuit of knowledge by scientists-citizens in a self-governed republic of science, free of control and regulation. The theme was continued by scientists and scholars whom Shils invited to respond to the opinions expressed by Weinberg and Polanyi—"the *Minerva* debate on scientific choice" as one of the participants, philosopher Stephen Toulmin, called it.²⁴ In his review summarizing the "*Minerva* debate" Toulmin concluded that Polanyi's image of the "republic of science" was out of date:

In real life, the republic of science cannot stand apart from the general commonwealth. Back in the 1930s, Polanyi's campaign to defend the autonomy of science against projects for a Nosey-Parkerish state centralism had a real point. By the 1960s, the need for academic science to be self-governing seems to be being conceded even in Russia and Polanyi's protestations are—surely—more insistent than they need be. As the social sciences too approach their coming-of-age, his distinction between the republic of science and the rest of the community becomes excessively disjunctive. The urgent question today is, rather, how the self-governing republic of science is to be integrated, not only into the broader academic confederation, but into the whole community of citizens.²⁵

Polanyi responded to the critique without, however, changing his general position. In "The Growth of Science in Society," published in *Minerva* in 1967, he reflected on the relations between science and society in the age of Big Science. Polanyi argued that Big Science doesn't challenge the science-as-market model. The market-based economic model was still valid, even in seemingly non-market situations: "The marginal principle of economics offers the conceptual model for [scientific research]. ... Funds and appointments serving scientific research must be distributed in a way that promises the highest total increment to science. ... The two great principles of scientific growth [are] the granting of independence to mature scientists and the imposition of scientific values on their performances ... achieved by ... *self-coordination by mutual adjustment and discipline under mutual authority*."²⁶

Weinberg did not find this view appealing. In 1968, reviewing a book by the science journalist Daniel Greenberg with the telling title *The Politics of Pure Science*, in which Greenberg contested the "traditional scientific view" that the internal system of science is a sufficient guarantee to require no outside surveillance, control or reinforcement, Weinberg wrote:

This traditional view has been articulated eloquently in the pages of *Minerva* by Professor Polanyi. In Polanyi's "republic of science" countervailing "political" forces are kept in rational equilibrium by the internal workings of the value system of science, a value system the ultimate sanction of

which is nature itself. Yet, on this issue—the effectiveness of the republic of science in keeping the game honest—I find myself more in sympathy with the journalist Greenberg than with the scientist Polanyi.²⁷

In 1991, Weinberg was even more explicit in his assessment of his own position, writing about his advocacy of Big Science as "being organized, more or less by an intrusive government":

This "socialist" view of science contrasts with Polanyi's Republic of Science, in which myriad independent practitioners determine the course of science. The Republic of Science is a free market and decentralized. My scientific enterprise is more socialist and centralized. Actually, I would say that where Polanyi's democratic republic is a good model for Little Science, my socialist republic applies more to Big Science.²⁸

And Weinberg added another twist to "*Minerva* debate." The choices rationalizing the decisions pertaining to Big Science, Weinberg argued, should be made not by politicians but by "some well-informed observers"—experts not in sciences per se but rather in meta-studies of science. "For this reason alone philosophic debate on the problems of scientific choice should lead to a more rational allocation of funds."²⁹ Big Science, in other words, required what might be called "Big Science studies"—an independent and decentralized expertise, which would provide a systematic study of Big Science mode of research and advise the government accordingly.³⁰ Weinberg jokingly remarked, in retrospect, that his papers in *Minerva* had launched his career as a "moonlight philosopher of scientific administration," and the "*Minerva* debate" had contributed to the recognition of "the importance of philosophic examination of the sanctions for public support of science," stimulating "something of a cottage industry in the philosophy of science policy."³¹

What Weinberg called a "cottage industry in the philosophy of science policy" had originated earlier, however. The founding of *Minerva* in 1962 had been preceded by three years of activities and workshops on science policy and science politics that Shils had organized under the auspices of the Congress of Cultural Freedom (CCF). The examination of the phenomenon later called Big Science was at the center of the intellectual agenda of the CCF. Both Shils and Polanyi were active members of the CCF and major drivers of the CCF Seminar Program and its "Study Groups," focused, among other areas of concerns, on the discussion of science as a social and political institution. The quest for "expertise" in meta-studies of science that Weinberg envisioned in his essays was at the center of the CCF's own aims and aspirations.

The Congress for Cultural Freedom and Its "Study Groups"

The Congress for Cultural Freedom is remembered as the spearhead of Cold War American cultural diplomacy, established with the aim to secure support for the Marshall Plan (Economic Recovery Program) in the sphere of culture and ideas by building

a transnational network of anti-Stalinist scholars and establishing strong bonds between European and American intellectuals. The CCF's goals were to be achieved through widely publicized endeavors such as the Festival of the Twentieth Century, as well as through CCF-affiliated journals such as *Encounter*, *Preuves*, *Der Monat*, and, later, *Minerva*.³²

The CCF's first visionary was Arthur Koestler, a Hungarian émigré whose widely known novels denounced totalitarianism and voiced calls to arms for the defense of freedom and the fight against Stalinism. Koestler's "militant liberalism" wasn't embraced, however, by other visionaries of the CCF. Thus, Nicolas Nabokov, Secretary General of the CCF, referring to Koestler's original plan for the CCF, which he envisioned as a "combat unit" devoted to the cause of the defense of freedom, expressed the general opinion in 1952: "the Congress [should be established] in the minds of the European intellectuals as a positive, and not only as a political, organization."³³ After Stalin's death, it became even more obvious that the simple message of "anti-communism" wasn't enough to sustain intellectual appeal and gather significant audience among the European and American intellectual elite.³⁴

The CCF lost no time establishing its "positive program," which emerged in the mid 1950s from the network of intellectuals associated with the CCF, most prominent among them Raymond Aron, Daniel Bell, and Edward Shils. The CCF's "positive program" was encapsulated in the rhetoric that later found its sharpest articulation in Bell's 1960 book *The End of Ideology*. Within the CCF, the "End of Ideology slogan" was adopted, in words of Michael Polanyi, as "an expression of [the CCF's] predominant aims [and] our official pronouncement."³⁵

The "end of ideology" rhetoric signified an important shift in the CCF's self-perception, "from an instrument of struggle against totalitarianism to an international forum for debate."³⁶ In Polanyi's words, the "end of ideology" was a "distinctive, passionately sober approach to culture and politics" that encouraged factual and calm examination of political systems and contemporary societal phenomena, rather than mere denunciation of totalitarian regimes.³⁷ This rhetoric sought to move away from simplistic early Cold War dichotomist imageries of incommensurable Eastern and Western world systems and competition promulgating instead the images of convergence and compromise. It promoted the view that the common forces resulting from the dramatic advances of science and technology were leading both Western capitalism and the Soviet Bloc's socialist and collectivist systems to adopt similar methods of socioeconomic management. In words of Konstantin Jelenski, a Polish émigré writer who led the Eastern European Division at the CCF, there was a "growing realization that the realities of industrialism are perhaps a more important determining factor socially than political systems, whatever their ideological origin."³⁸

The "end of ideology" wasn't merely a normative position, it sought to offer a substantial reformulation of the ideals and goals of classical liberalism. In Polanyi's

words, the "end of ideology" was the means to "secure a post-Marxian basis for liberalism throughout the world."³⁹ These realities of the postwar world had shaken the belief in an unregulated free market system and *laissez-faire*. The "end of ideology" presented a renewed defense of capitalist "free society," with its central ideals of the "free market" and *laissez-faire* economics, seeking to attune it to the socio-political implications of changes in political economy in the postwar industrialized world, and to the subtleties and compromises of the Cold War with its postwar coalitions.

With this agenda, the "end of ideology" rhetoric emphasized the studies of science as the topic of central concern. With its emphasis on "sober," sophisticated, dispassionate socioeconomic analysis of modern industrial societies and their political systems, the "end of ideology" promoted the view that a dramatically increased role of science in the realm of public affairs and politics had delineated a new phase in the development of Western liberal democracies. Science, its history, and its politics were to be assessed, especially in terms of their implications for democracy, liberalism, and freedom. The CCF intellectuals sought to offer such an assessment, and they did this in a big way.

The CCF "methodology": The Study Groups and the Seminar Program

As Rebecca Lemov has pointed out, "methodological thought" became a preeminent concern during the Cold War.⁴⁰ Within the CCF this certainly was the case. With its "end of ideology" rhetoric calling for a non-ideological "sober" research based on facts, rational method, and science, rather than ideological conceptions and "messianic claims," the CCF encouraged the application of scientific methods to social problems. Among the models for the CCF's activities was the Committee for Economic Development (CED), which was seen by the CCF leaders as an exemplary organization that had developed sophisticated methods of collecting, processing and analyzing vast amounts of social data:

The Congress [for Cultural Freedom] would do well to adopt this CED technique or some appropriate variant of it whereby data which has been collected by specialized institutions on different topics of general importance would be submitted to a small study group of the Congress which would prepare on the basis of the detailed examination of the available data, recommendations for a special international policy statements, to be issued on the name of the Congress. Such statements ... could conceivably have a beneficial influence on the policies of other organizations, of foundations, and even of governments.⁴¹

The CED model of quantitative analysis of large sets of social data was, however, soon abandoned in favor of a qualitative analysis. The main form of the CCF's activities became the CCF Seminar Program, conceived as an innovative form of interdisciplinary contact, a method of "intellectual confrontation," in which critical discussion was methodically staged. One of the CCF officers claimed that "the seminar concept ...

provides the most adequate form for the treatment of many issues ... stimulating thinking and discussion about ... new ways of organizing intellectual confrontation."⁴² To achieve this healthy "intellectual confrontation," the CCF's seminars and study groups meetings adopted the practice of commissioning two or more position papers presenting opposing views "on a critical contemporary problem" (for example, "an optimistic and pessimistic view of the quality of life in industrial society") and submitting them for discussion in small groups.⁴³

The CCF Seminar Program consisted of international seminars and a large number of smaller local meetings. The themes were diverse. Science and its role in the postwar world was but one of many topics, but it always loomed large on the agenda.⁴⁴ The unstated goal of the Seminar Program was to develop the CCF into a clearing house for independent expertise on the pressing issues of the day—expertise that would eventually be used to advise government policy makers. The discussions that took place, which were carefully recorded and transcribed, were ultimately intended to result in a series of publications devoted to "major present preoccupations of the intellectual world."⁴⁵

Edward Shils, one of the chief designers of the CCF's seminar program, also helped to introduce another form of the CCF activities: study groups. CCF study groups originated in the "Study Group on Science and Freedom," organized by Michael Polanyi as part of the 1953 Hamburg Congress and an outlet of his Society for Freedom in Science that he formed in the 1940s.⁴⁶ For Polanyi, the system of the organization of science as it had developed since the Scientific Revolution was a resource for liberal capitalist democracy.⁴⁷ The discussion of these ideas at 1953 Hamburg Congress resulted in Shils' proposal to set up a Study Group within the Congress that would sharpen the picture of the relations of science and the state in the contemporary world.⁴⁸

At the 1955 conference in Milan, Shils suggested that the Study Group be extended into a "future Congress on Science and Freedom" that would focus on the "impact on the academic community of the great historical changes of the past two decades, and particularly of the greatly increased need for finance from outside sources, and the invasion of practical and technical tasks in the academic sphere, with all its resulting social, political and intellectual problems."⁴⁹ The envisioned Congress, eventually held in Hamburg in 1959, was devoted entirely to discussion of the dramatic changes in the role of technology and science that had occurred since World War II. The aspiration was to produce a comprehensive account of the "Technical Age" into which both Western and non-Western societies had entered.⁵⁰ However, the topic was obviously too grandiose and too important to be limited to one meeting. As a solution, Shils suggested dividing it among several permanent Study Groups. The CCF adopted Shils' proposal, and six Study Groups were inaugurated, each having its own program within the general theme outlined at the 1959 Hamburg Congress. The Study Groups continued to function thereafter, with different levels of success and productivity.

As Shils outlined the "division of labor," one Study Group, directed by Shils himself, would focus on the political aspects of the "Technical Age," characterized by the "decline of ideologies" in Western societies and the rise of technical and scientific expertise with their growing role in the realm of public affairs. Two other groups would be concerned with the features, both positive and negative, of modern industrial and mass society, with one group focusing on the features of society while the other focused on the individual's relation to culture and society in the "Technical Age." Then, two groups would focus on the role of the "vast class of intellectuals"—artists, scientists, and academics of all kind—in the contemporary world divided by the Cold War. Finally, the last group would study "the signs already visible of affirmation in the Communist world itself of those values which we hold dear. It will seek to pinpoint indications that such values do provide a point of convergence for various contrasting historical developments."⁵¹ The plan was adopted by the Study Groups' "directors"—Edward Shils, Michael Polanyi, Daniel Bell, Raymond Aron, and Nicolas Nabokov. In subsequent years, the topics of Study Groups changed, although there always was a continuity with the original themes suggested by Shils. Shils, as a Study Group leader, was interested in science policy and science politics; Polanyi was concerned with the relationship of thought, mentalities, and contemporary politics; Raymond Aron led the discussions of the conditions of stability, both in democracies and ideological orthodoxies; Nicolas Nabokov led the discussions of the role of the arts in the postwar world.⁵²

In 1960, Shils focused his Study Group on "Scientific Policy—the Cooperation of Government, Economy and the Universities in the Development and Application of Scientific Research."⁵³ That year and the next, he invited leading academics and scientists to take part in discussions, or to contribute a paper or a statement to be discussed. As one of the founders of the *Bulletin of the Atomic Scientists*, and as a co-founder and vice-director of the University of Chicago's Office of Enquiry into the Social Aspects of Atomic Energy, Shils had long been involved in the atomic scientists' movement. Now he capitalized on this network, trying to enlist atomic scientists in a more general and theoretical discussion of the interactions between science and society. "The kind of persons we wish for," he told J. Robert Oppenheimer in 1960, "are scientists concerned with problems of scientific policy, 'scientific administrators,' civil servants and politicians especially concerned with the development of science and other aspects of the application of scientific knowledge."⁵⁴

By 1962, Shils was ready to move his Study Group's discussions into the public realm. The CCF Study Group on Science Policy was transformed into the CCF-affiliated journal *Minerva*. Establishing *Minerva*, Shils transferred some of the approaches and formats of the CCF Study Groups workshops and discussions into this forum. For the opening issue of *Minerva*, in line with the methodology of discussion that was adopted at the CCF, Shils commissioned two opposing position papers on Big Science: an

“optimistic” one, by the physicist and science administrator Alvin Weinberg, and a “pessimistic” one, by the physical chemist and philosopher Michael Polanyi. Unlike the CCF seminar program, however, *Minerva* was established as a public forum with the intention of influencing not only the intellectual elite (the major “target” of the CCF’s activities) but also the broader academic community and the makers of science policy.

In the 1960s the CCF sponsored a number of influential magazines and organized large and small international conferences and seminars on a wide range of topics, including science and its roles in the broader culture, in society, and in politics. By the mid 1960s, the CCF was regarded as a big success. Referring to the Congress’ role in countering the appeal of the pro-communist Left and in creating a receptive atmosphere for the formation and support of a movement that often thought of itself as on the anti-communist Left, one of the CCF’s best known members, the diplomat and the State Department’s Soviet expert George Kennan, wrote in 1959 to Nicolas Nabokov: “I can think of no group of people who have done more to hold our world together in these last years than you and your colleagues. In this country [i.e. the United States] in particular, few will ever understand the dimensions and the significance of your accomplishments.”⁵⁵

“Who paid the piper”: The CIA connection and the moral crusade of the CCF

In 1967, a series of newspaper publications revealed the close association of the CCF with the US Central Intelligence Agency. The CIA connection presented an ultimate—and very sensible—test for the theoretical discussion of the societal effects of the “governmentalisation of science” and the issue of “the indirect as well as the direct influence of the Cold War on the role of science and scholarship” (quoting Shils’ 1962 editorial). During the 1960s, the revelations of the CIA’s infamous involvement in operations in Iran and Guatemala in the 1950s, and in Cuba and Vietnam in the 1960s, made the question of “governmentalisation of science” as much a moral as an epistemological position for the CCF-associated intellectuals and for their critics.⁵⁶

The revelation of the CIA funding behind the CCF and its activities caused public outrage. What can a “free thinker” say about his “freedom,” asked the *Sunday Times* of London, “when he finds out that his free thought has been subsidized by a ruthlessly aggressive intelligence agency as part of the international Cold War?”⁵⁷ A report in the *New York Times* similarly pointed out that the intellectuals who were funded by the CIA (with or without their knowledge) were “being used for concealed government propaganda,” which made a “mockery” of intellectual freedom. In the aftermath of the controversy, the historian Christopher Lasch commented that “the whole wretched business seemed inescapably to point to the conclusion that cultural freedom had been consistently confused with American propaganda, and that ‘cultural freedom,’ as defined by its leading defenders, was—to put it bluntly—a hoax.”⁵⁸

For most of the CCF associates, the revelation about the Congress' link to the CIA didn't come as a surprise. As Sidney Hook admitted retrospectively in his autobiography, "I have heard, like almost everyone else, that in some way the CIA was involved in funding the congress. Everyone mentioned it, even though no one had any hard evidence. ... In my own mind I had no doubt that the CIA was making some contribution to the financing of the Congress. ... Everyone involved in the activities of the Congress had heard rumors of covert CIA support."⁵⁹ Most of the CCF's members continued to be engaged in the organization, either knowing or suspecting its source of funding, assuming that as long as they are not dictated or controlled in their intellectual activity, they can claim their intellectual independence and integrity. Within few days after the revelations in the *New York Times*, several distinguished CCF associates—John K. Galbraith, George Kennan, J. Robert Oppenheimer, and Arthur Schlesinger Jr.—wrote a letter to the editors of *New York Times* stating: "on the basis of our own experiences with the Congress over the past 16 years—with its seminars, its artistic festivals, its magazines, its staff—we can say categorically that we have no question regarding the independence of its policy, the integrity of its officials, or the value of its contribution. In our experience the Congress ... has been an entirely free body, responsive only to the wishes of its members and collaborators."⁶⁰

Neither *Minerva* nor its editor was directly attacked in the press, but Edward Shils was as disturbed as others when he saw that their valuable enterprise was being discredited. Like other CCF-associated intellectuals, Shils insisted that the CCF's magazine's editorial independence wasn't corrupted, emphasizing his own loyalty only to the "commitment to cultural freedom."⁶¹ Deploying the "end of ideology" rhetoric and posture, Shils went so far as claiming that the CCF wasn't "political." In a letter to Crawford Goodwin, a professor of economics at Duke University and the program officer in charge of European and International Affairs at the Ford Foundation, he wrote: "it might be reasonably claimed that the Congress ... was not political. It sought to promote the understanding and solution of fundamental problems which concern serious intellectuals ... cutting across the boundaries of nationality, party, intellectual field and discipline. ... It created and fostered a sense of affinity among these intellectuals in a way which is, I think, unique in the history of the present century."⁶²

The CCF's leaders should have felt that they were between a rock and a hard place. The complexities of the interrelation between politics and science that they had been discussing at length in the seminars and study groups appeared to represent now their own predicament. In some ways, the CCF intellectuals were reasoning according to the very logic of "Big Science" that they were disentangling. They were accepting, with Weinberg, that Big Science had changed not only science but also the way the relation between science and the state was understood. If Big Science was political

then “Big Scientists” could not but be politicians in their claims on freedom and independence of science vis-à-vis state and politics.

The positions taken in the “*Minerva* debate” continued to resurface within the network of the CCF intellectuals, now on the “moral plane.” The “intellectual confrontation” staged by Shils on the pages of *Minerva* resurfaced as a “moral confrontation” among the CCF leaders now facing a moral conundrum. Polanyi, once again, presented a “dissenting” view. Reasoning consistently with his theoretical defense of “Little Science,” Polanyi was much concerned about a “little person”: Michael Josselson, the CCF’s Executive Director and a dedicated driving force behind the CCF’s activities since its inception. After he was revealed to be a CIA officer, Josselson was ostracized by other CCF leaders, who unequivocally insisted on his immediate resignation. Polanyi, alluding to Arthur Koestler’s famous novel, called the decision to “eliminate” Josselson from the CCF “another darkness at noon.” For Polanyi this decision was, first of all, morally wrong. Moreover, it wasn’t “rational.” It didn’t resolve the moral conundrum; it was only a way to claim unawareness about the CCF’s connection to the CIA, a position that Polanyi found neither moral nor rational. In a letter, he asked Raymond Aron “What kind of figure are we going to cut? Are men like you or me ... going to declare that in 15 years we did not notice that we were being manipulated to serve sinister purposes? Are we going to proclaim our awakening, a new version of *The God That Failed*?”⁶³

The fate of their “valuable enterprise” was, in Polanyi’s eyes, inseparable from the fate of its “little” leader, even if it would imply to “lose face” and admit the strong connection between the CCF and the CIA. Appealing to the CCF’s support “for saving Mike [Josselson], the Congress, and our honour,” Polanyi wrote the following to Aron:

I would have served the C.I.A. (had I known of its existence) in the years following the war, with pleasure. We were faced with an ubiquitous madness, supported by an empire and organized on conspiratorial lines. ... In the years after 1950 we battled against a phalanx of Stalinist or Stalinisant intellectuals throughout Europe, for the vindication of free thought, which was despised and ridiculed by those who are now forcing us to dismiss Mike Josselson, because he had accepted the support of like-minded American officials, who appreciated the ideals he was fighting for.⁶⁴

In the end, it was the forced resignation of Josselson rather than the revelation of CIA funding that prompted Polanyi to resign from the CCF. As he explained to Pierre Emmanuel, “I expressed the feeling that I could not remain connected with our organization, if we decided to eliminate Michael Josselson from it. ... I beg you, therefore, to accept my resignation from the community to which I have so long adhered.”⁶⁵

With funding secured by the grant from the Ford Foundation, the CCF quietly became the International Association for Cultural Freedom.⁶⁶ The IACF “inherited” the CCF’s magazines and, aside from a few resignations, continued to rely on old

networks until it quietly dissolved itself in 1979, never having enjoyed as much success as the CCF had enjoyed throughout the 1960s.⁶⁷

The Broader Context of the "*Minerva Debate*": The Theories of "Post-Industrial Society"

The "*Minerva debate*" articulated the awareness that the growth of science, for better or for worse, had significant implications for modern society. By the 1970s, this view had become common. Jacques Barzun, Spencer Klaw, Edward Shils, Daniel Bell, John K. Galbraith, and many other social theorists promoted the view that access to the power of the atom, the computer revolution, the exploration of the cosmos, and the greater cultural, social, economic, and political significance that science had come to hold in Western societies delineated a new type of social order—a "post-industrial society."

The theory of "post-industrial society" was the most articulate descriptor of a new postwar social structure in which advanced science and technology played a major role. Like the "end of ideology," the theory of "post-industrial society" was formulated within the CCF's network of intellectuals. It was most prominently articulated by Daniel Bell in his 1973 book *The Coming of Post-Industrial Society*.⁶⁸ However, the concept was born much earlier, within Bell's Study Group and the CCF seminars.

In his discussion of the changing role of the intellectuals, Bell attributed central importance to the technological developments that transform and increase the number of scientists and men of knowledge, seeing them as the dominant element within the new social structure in a society where the social functions of science had considerably expanded. The post-industrial society was a society in which the growing role of intellectual activities implied not only the mobilization of science for the sake of both knowledge and capital production, but also suggested that the distinction between economic and social processes were increasingly vanished. In post-industrial society, new forms of community emerged as alternatives to market-based norms of organization, and a "professional and technical class" of scientific workers displaced the "old working class."

The "post-industrial society" rhetoric resonated with the changing context of the Cold War. By the late 1960s, a partial *détente* with the Soviet Union and the "communist" countries in Europe made the anti-communist rhetoric of the 1950s obsolete, though of course didn't make anti-communism obsolete. The anti-communism of the late 1960s and the 1970s required a new rhetoric, and the "post-industrial society theory" supplied exactly what was needed, presenting the images of the emergence of a unitary "post-industrial society" in both mature capitalist and socialist "techno-structures"—the so-called "convergence theories," widely promulgated throughout the 1970s. As Bell affirmed, behind the argument about convergence lay a recognition of the fact that, in Bell's words, "the market was rediscovered in the Communist socialist world, and the market was losing its importance in the Western economies."⁶⁹

Economic performance and its prerequisites, scientific and technological advances, were the defining elements of modern life and a measure of success in the Cold War.

The “*Minerva* debate” provides a perfect illustration of how the “planning-versus-market” dichotomy didn’t hold in the world of Big Science shaped by Cold War. The Western democracies and the Soviet Bloc were claimed to represent two incommensurable and opposing forms of economic modernity, epitomized in two ideal types—Soviet *planning* and American *free market*. Yet, in the debates on the pages of *Minerva* these two ideal types were redefined, re-negotiated, and reconciled to accommodate actual practices of social planning and control existing on both sides of the Iron Curtain. As a result, the “planning-versus-market” dichotomy was broken down into a number of different positions across the entire spectrum from planning to market.

Shils, eager to include the perspective from the Soviet side, published several papers in which the physicist Pyotr Kapitsa discussed the planning of science in the Soviet Union. The papers by Kapitsa published in *Minerva* were translations of articles that had been published in *Pravda* and in *Literaturnaia Gazeta* in 1966.⁷⁰ Kapitsa argued that Soviet science, which had been planned at the state level for several decades, had reached the point where a more effective approach to research should be implemented—one that would take the views of scientists into account. Emphasizing the value of the creative element in science, Kapitsa argued that “the plan” should support rather than restrict the freedom of scientific creativity and should be implemented by scientists “free[d] from petty controls.”⁷¹

Kapitsa’s deliberations came at the time when the prestige and authority of scientists had risen dramatically in the Soviet Union, and Soviet physicists used this power and prestige to renegotiate their relations with the party-state and to launch an active campaign for a liberal reform in science policy.⁷² Referring to the advocates of the view that science should be left to the scientists in the Soviet Union, Kapitsa wrote: “The scientists who maintain that scientific work cannot be planned because it develops spontaneously are wrong”⁷³ Instead, Kapitsa advocated a more moderate position, arguing for a reform of the system of planning of science that would allow for more initiative on the part of scientific institutions and scientists themselves. In this way, Kapitsa contended, Soviet science could take advantage of the “freedom of science,” a cherished value of Western scientists, as well as of the strength of Soviet institutional system of support. “We have money for research,” Kapitsa noted. “[T]he state is not mean and we can get it more easily than, for instance, American scientists.”⁷⁴

Kapitsa’s articles, in Shils’ view, resonated with the consensus articulated during the “*Minerva* debate.” Indeed, the reasoning of a representative of Soviet scientific community matched the tone of the discussion by American scientists and scholars trying to come to terms with Big Science’s centralization and planning coming along with generous federal and military patronage. Sending a translation of one of Kapitsa’s

articles to his assistant, Shils wrote "This article is of first importance. It must get as much publicity as possible when appears."⁷⁵

The publication of Kapitsa's essays in *Minerva* also marked a watershed in the way Soviet experience was presented in discussions of the interrelation of science and politics by American scientists after World War II. In the early 1950s, the "Lysenko affair" and the scandalous ban of genetics in the Soviet Union in August of 1948 became the most forceful arguments in favor of the "autonomy" of science and *laissez-faire*. Thus, in 1953, at the CCF's Science and Freedom conference in Hamburg, Polanyi forcefully argued against attempts by John D. Bernal and other left-wing British scientists to promote Soviet science policy as an appropriate model for the West. Polanyi backed his argument against planning of science by reminding listeners of the damage that had been done by Lysenko's misconceived effort to "run science for the public good."⁷⁶ By the early 1960s, however, in the context of the debate on Big Science on the pages of *Minerva*, Lysenko wasn't in the picture anymore. In the early 1960s the Soviet Union was seldom described as failing. Its economy tended to grow faster than that of the United States, and especially after the launch of Sputnik in 1957 the Soviet slogan of "catching up and overtaking" the West didn't appear to be patent nonsense.⁷⁷ In the 1960s and the 1970s, a more important lesson from the Lysenko affair was the acknowledgment that the state planning of science may produce different results depending on the nature of the state.

How were all these issues seen on the other side of the Iron Curtain? The theoretical discussions of the impact of the growth of science on society in the postwar world weren't restricted to the West. Simultaneously with the debate on "Big Science" in the United States, a parallel discussion of what was called the "Scientific-Technological Revolution" was taking place in the Soviet Union. Like their American counterparts, Soviet scientists and scholars responded to the anxieties and concerns of the Cold War, adapting and transforming them in highly specific and often peculiar ways.

The Soviet "Counterpart" of Big Science: The Theory of "Scientific-Technological Revolution"

In the late 1950s and the 1960s, in the wake of the sensational achievements of Soviet nuclear physics, the construction of the first nuclear power station and the first nuclear-powered ice breaker, and, above all, the Soviet space-exploration program, a new term entered the Soviet political and philosophical lexicon: "Scientific-Technological Revolution" (Nauchno-Tekhnicheskaiia Revoliutsiia).⁷⁸ It came to denote the postwar scientific achievements and technological innovations that had taken place since World War II and that promised to change the socioeconomic conditions in the USSR and other countries.⁷⁹

A comprehensive theory of Scientific-Technological Revolution (hereafter abbreviated to STR) was first developed during the 1960s by philosophers and social theorists in East Germany and Czechoslovakia, the most technologically advanced countries in the Soviet Bloc. In Czechoslovakia, the concept of the STR, developed by Czech philosophers, provided the Czech reform movement led by Alexander Dubček with its philosophical agenda. The program of Czech economic reforms ("socialism with a human face") was the result of teamwork by philosophers, economists, sociologists, psychologists, engineers, and natural scientists commissioned by Dubček's government and headed by the philosopher Radovan Richta, who assumed the directorship of the Institute of Philosophy in Prague in 1968. A programmatic collective monograph, *Civilization at the Crossroads: Social and Human Implications of the Scientific and Technological Revolution*, was published in Czech in 1967 and in English in 1969.⁸⁰ The central argument of Czech reformers was that the STR marked a new epoch "in the evolution of productive forces" and required the adaptation of the socialist economic system to the demands of modern industrialization and scientific-technological development. The STR was critically different from the "first industrial revolution" in many important aspects, Czech reformers argued, because in the socialist countries the qualitatively new possibilities of the STR were combined with an economic system inherited from the first industrial revolution. "These considerations," according to Richta's report, "underscore the vital need for radical economic reforms now being introduced in the socialist countries."⁸¹ In practice, such an adaptation would require a more flexible and transparent economic system, and openness to the worldwide exchange of information and ideas.

The Soviet invasion of Czechoslovakia in August of 1968 and the end of Dubček's "new way toward socialism" had immediate consequences for philosophical discussion of the STR. Czech specialists who played leading roles in the debates of the 1960s, particularly those who were active in the reform movement, lost their formal positions, although some continued to work in less visible roles as researchers. Some theoretical innovations of Czech reformers, particularly concerning the role of social groups and the nature of social relations under socialism, were explicitly rejected.⁸² The major effect of the events of 1968 on the theories of the STR was a shifting of the focus from the reformist cause (that is, the need for socialist societies to adjust to the requirements of the STR) to discussion and evaluation of the STR's anticipated or unanticipated social and political consequences.

The 1970s marked the beginning of what might be called the "era of the STR" in Soviet political discourse, when the STR became a central notion in the official statements of Soviet leaders. The greatly increased official commentary on the STR was part of the campaign to formulate national policies and to mobilize bureaucratic support for the major economic and political decisions of the late 1960s and the early 1970s. Nixon's visit to Moscow in 1972, followed by the accords on Soviet-American

cooperation in space exploration, environmental matters, and trade, marked the beginning of détente, which was presented in official Soviet pronouncements as the end of the Cold War. Rather than ending the Cold War, however, détente introduced a new phase of it, marked by increased economic cooperation and trade.

The extension of foreign trade and the importation of Western technology in the Soviet Union had already increased greatly under Khrushchev, after the domestic liberalization and opening up of the Soviet economy to the outside world after Stalin's death. Although in the late 1950s and the 1960s the Soviet economy was growing faster than American one, with Soviet technological confidence boosted by the successful launching of Sputnik in 1957 and by other achievements in space exploration and in nuclear technologies, top Soviet officials acknowledged major weaknesses in domestic research and development. The economic historian Philip Hanson traced the beginning of the internal debate on the economic reforms within the Party Central Committee to 1958, when a Plenum of the Central Committee encouraged systematic attempts to make the planning of Soviet R&D more comprehensive.⁸³

The oil crisis of 1973 created new opportunities for the exporting of oil and opened new possibilities for foreign purchases and restructuring of Soviet economy. The steep rise in oil prices radically improved the Soviet Union's terms of trade and gave an enormous boost to the foreign purchases, leading Brezhnev's government to adopt a long-term policy of technology transfer from the highly industrialized capitalist West.⁸⁴ During the period 1965–1972, two huge vehicle manufacturing plants, one for cars (the Fiat-Togliatti plant) and one for trucks (the Kama River plant), were constructed by purchasing equipment and general technical services from Italian and American companies.⁸⁵ The Kama River complex was parceled out to a number of different Western contractors, since no single Western company was prepared to take the risk of being the general contractor.⁸⁶ In both cases, the whole systems of technological know-how were transplanted from capitalist West to socialist East. Thus, during the construction of Fiat-Togliatti plant in what is now the province of Samara, Western contractors had overall responsibility for designing, equipping, installing, and commissioning the production facilities. In addition, they trained the Soviet workers, which resulted in significant cross-border flow of machines and people. Western specialists flew to the construction site and to Moscow, where the design bureau was located; Soviet citizens flew to Italy to learn how to operate the equipment that was being installed.

The emphasis of Brezhnev's government on the importation of whole systems of technology from the capitalist West made theories of the STR the main element in evaluating the effects (especially undesired and unanticipated ones) of the infusion of foreign technology into socially and politically different societies. In this context, Soviet STR theorists were expected to offer a comprehensive discussion of the relations between technology and society. They did as well as they could within the constraints

of the official philosophical discourse. Questioning conventional interpretations of the exchange of technology as a form of cultural diffusion, they postulated analytical distinctions between the form and the content of modern technologies, between “technique” (gadgets and machines) and “technology” (social and economic relationships embedded in supposedly value-free machines), and between direct and indirect effects of technology on society. In a sharp contrast with usually highly abstract philosophizing, this discussion was tied to the real-life situations and processes. For example, in 1972 the philosopher Genrich Volkov contended that some technological innovations, for example computer technologies, increased labor productivity indirectly through the changes in production relations, while other technologies had a direct effect on labor process, “relatively independent of socio-economic operations.” Thus, Volkov concluded, “an assembly line would require the same type of highly specialized, mechanical operations, no matter whether it is installed in a Detroit plant or in a plant in Sverdlovsk.”⁸⁷

These discussions revealed deep disagreements about the social and political implications of technology transfer. The authors of a 1972 volume titled *Nauchno-tekhnicheskaiia revoliutsiia i social'nyi progress* contended that new technical hardware couldn't simply be grafted onto existing processes of labor, production, and management; rather, the existing processes required changes in order to accommodate new machines. Technical breakthroughs could be utilized effectively only if adjustments were made to the larger social systems. For example, the installation of computers would not produce “revolutionary” changes in the forms and organization of production, or in decision-making practices, unless they were accompanied by changes in the organization of the flow and content of technical and social information.⁸⁸ The underlying argument was that techniques could be considered value-free, but they were embedded in the value-laden technologies that enabled them to perform social and political functions.

The major producer of literature on the STR in the 1970s was the Institute for the History of Science and Technology (Institut Istorii Estestvoznaniia i Tekhniki) in Moscow. The IHST traces its origin to the Institute for the History of Science established under the directorship of Nikolai Bukharin in 1927, which was disbanded after the arrest of Bukharin and then that of another short-tenure director in 1938 during the Stalinist purges.⁸⁹ In 1945 the institute was reestablished with a new goal—to provide historical evidence to the Soviet nationalistic campaign in the wake of World War II. In the early years of the Cold War, the main function of the IHST was to assert the superiority of Soviet science.⁹⁰ In 1962, with the appointment of the new director, the philosopher Bonifatii M. Kedrov, the profile of the IHST changed dramatically.⁹¹ Kedrov's program included a strong philosophical component and established strong connections between two Moscow institutes, the Institute for the History of Science and the Institute of Philosophy. One of the IHST's newly formed departments,

dedicated to "general problems of the history of science," was staffed by philosophers from the Institute of Philosophy, who were expected to provide direction and theoretical framework to historians of specialized branches of science.

The "philosophical turn" of a historians' Institute served several goals. One of these goals concerned the new role for the history of science, as it was redefined in the 1960s. During the nationalistic campaigns of the late 1940s and the 1950s, the IHST dutifully produced or supervised numerous works that claimed priority for Russian science in almost any significant scientific discovery and trumpeted the superiority of Soviet science system over the Western one. At the same time, as a result of these militant nationalistic campaigns many historians retreated to descriptive and antiquarian modes of scholarship, or to exotic fields such as Arabic mathematics and ancient science.⁹² In the 1960s, neither of these strategies of historians suited the new political agenda epitomized by the notion of the STR. The rhetoric of the STR had emphasized *recent* scientific developments and had encouraged study of the scientific and technological achievements of major *Western* industrial nations, rationalizing the new emphasis of Soviet economics on the transfer of Western technology and know-how. Kedrov's 1962 program emphasized that the IHST's structure and research focus should be changed to "correspond to the present day situation" and to ensure the production of up-to date "synthesizing and analytical work on the development of recent science and present-day science."⁹³ New departments—one dedicated to "the Scientific-Technological Revolution" and one to "General Problems of the History of Science"—were created to ensure the implementation of these changes.⁹⁴

During the 1970s, the IHST produced or supervised a large number of collectively authored books on the STR.⁹⁵ One of the most important was *Man-Science-Technology: A Marxist Analysis of the Scientific-Technological Revolution* (1973), written by members of two Moscow institutes—the Institute of Philosophy and IHST—teamed with the Czechoslovak Institute of Philosophy and Sociology in Prague, where Radovan Richta and a number of other authors of the Czech reforms manifesto *Civilization at the Crossroads* continued to work as researchers after the Prague Spring.⁹⁶ Whereas *Civilization at the Crossroads* had contained only few references to Marx and none to Lenin, and had cited primarily Western works dating back to the 1940s (for example, works by John D. Bernal), *Man-Science-Technology* tied the notion of STR to the classics of Marxism, though it placed no particular emphasis on Marxism-Leninism.

The malleable discourse of the STR served various ends. On the political level, theories of the STR were deployed both by conservative Party leaders (to justify and rationalize the preservation of the status quo) and by those who sought to rationalize "revolutionary" transformations in many areas of Soviet life by reducing organizational "irrationality" and "optimizing" economic decision making.⁹⁷ On a theoretical level, the STR represented a general theory of social change, and hence an important modernization of Soviet Marxism.⁹⁸

Whereas classic Marxist-Leninist theory of the 1930s had emphasized class conflict as the motor of social change, the basic premise of the STR was that the advancement of science and technology had become the principal driver of societal transformation. Unlike standard Soviet Marxism, this new outlook didn't take for granted that socialist society was the most advanced simply because of the public ownership of the means of production. Instead, the STR promoted a new image of competition between the two world systems based not on class conflict or military victory but on the Soviet Union's allegedly superior ability to develop, manage, and apply advances in science and technology.

For the Soviet philosophers, the theories of the STR presented a vantage point from which to create a new outlook and a modernized version of Marxism-Leninism, replacing the emphasis on the role of class struggle with the view that the ability to harvest the achievements of science and technology was a defining element of modern life. The theories of the STR extended Soviet Marxism-Leninism beyond the calcified form promulgated under Stalin. As the major evolving part of the official Marxist-Leninist theory in post-Stalinist Soviet Union, the STR discourse also restored, to a certain degree, the intellectual function to Soviet official philosophy, in the sense that political struggles were not solely over positions and personal power but also over ideas and the meaning of Marxism-Leninism.

The theories of the STR were conceived in dialogue with and as a response to the writings of American social analysts evaluating and forecasting the effects of large-scale growth of scientific research on society in the atomic and post-atomic age.⁹⁹ The theories of "post-industrial society" and "convergence theory" were thoroughly referenced, debated, and reviewed in the literature on the STR. In contrast to Daniel Bell and other theorists of "post-industrial society," Soviet analysts largely denied the emergence of a unitary "post-industrial society" in both mature capitalist and socialist systems.¹⁰⁰ Yet they presented an array of different opinions on social and political consequences of the STR. The opinions ranged from the assertion that the STR would result in a transition to "the STR society" and the convergence of the two systems into a unitary and international "industrial communism" to the view that the STR was connected not with convergence but with a divergence between the capitalist and Communist social systems and with the expectation of an "intensified ideological struggle."¹⁰¹

The bottom line of these diverse opinions was a cautious recognition that some of the *problems* of capitalist and socialist societies, in their drive toward an advanced stage of modernization, could be, indeed, common, but the Soviet theorists insisted that the capitalist and socialist societies differed in their *responses* to these problems. Georgi Shakhnazarov, a political scientist and a Central Committee department official, summarized the question in his 1979 book *Fiasko Futurologii*:

To be sure, in practice all or nearly all social phenomena bear the imprint of the prevailing system and class ideology. Even such seemingly nonsocial phenomena as *technology* and *technique* are no exception. They, too, can [acquire specific characteristics] in distinctive social conditions, giving rise to distinctive social consequences. ... Still, we must not overlook the fact that [these characteristics derive] not from the intrinsic nature of *technology* and *technique* but from the method of applying them in concrete social conditions. The gadgets and technologies are per se products of the human brain and human labor. ... Their spread and the increasing resemblance of production processes is, therefore, no argument in favor of any convergence of the social systems. ... Past experiences have shown that similar problems may be resolved in quite different ways, depending on the social conditions and the aims sought by the prevailing political forces.¹⁰²

The theories of the STR was a Soviet counterpart to the American concept of Big Science. On the both sides of the Iron Curtain natural scientists and social theorists attempted to delineate a new epoch in human history based on the advancements of science and technology in the new, atomic or post-atomic, age. Both were articulated in response to the local economic and political situation and the needs of both states during the Cold War. In both political settings, reflection by natural scientists and social analysts on the social and political consequences of Big Science, as well as the articulation of the need for independent expertise on Big Science, was an important context for the nascent field of "science studies."¹⁰³ As the reception of Thomas Kuhn's work in the Soviet Union demonstrates, the Soviet "Big Science studiers" responded to the same anxieties and concerns of the Cold War as their Western counterparts, while adapting and transforming them in specific and often peculiar ways.

Reading Kuhn's *Structure* against the Backdrop of the Discussions on Big Science in America and the Soviet Union

Historical accounts of Science Studies (or Science and Technology Studies) usually highlight Thomas Kuhn's seminal book *The Structure of Scientific Revolutions* (1962) as challenging the received view of science and triggering the burgeoning of Science Studies as an academic field in subsequent decades. As David Hollinger has pointed out, however, many if not all of Kuhn's ideas can be found in the works of American intellectuals who were writing before or simultaneously with Kuhn.¹⁰⁴ Hollinger singled out Daniel Bell's *The End of Ideology*; Derek Price's *Little Science, Big Science*, Fritz Machlup's *The Production and Distribution of Knowledge in the United States*, Warren Hagestrom's *Scientific Community*, and Don K. Price's *The Scientific Estate* as constituting the immediate context of Kuhn's *Structure*.¹⁰⁵ Not only did those works emphasize the "communitarian aspect" of science; they also represented an important shift in the perception of science: from timeless and universal "science" without

national or political affiliations to science as a concrete, historical, and interacting community.¹⁰⁶

The “*Minerva* debate” on Big Science might be seen as part of this broader discussion that constituted the context of Kuhn’s *Structure*. Although the *notion* of Big Science is not found in *Structure*, the *phenomenon* of Big Science can be easily discerned in the background. Kuhn, a physicist turned historian, depicted the world he knew best—the practices and the political economy of physical science that overcame revolutionary changes in the wake of World War II. Kuhn’s scientists were team workers, “organization men” who followed instructions and defended their “paradigms.” These were the scientists of Big Science, not “little science.” However, these connotations were only implicit in Kuhn’s work.

Edward Shils sought to make it explicit. In 1962, Shils asked Stephen Toulmin to review of Kuhn’s *Structure* for *Minerva*. Indeed, the book came out in the midst of the discussions of the social and political implications of Big Science on the pages of *Minerva*. In his request, Shils made clear that he wanted to do some “translation work” for Kuhn’s *Structure*, relating it to the discussion of science policy and science politics for which *Minerva* provided a forum in the early 1960s: “[T]he main point about Kuhn is that it should be centered around the implication of Kuhn’s conception of scientific development for the planning and administration of science—not an easy task!”¹⁰⁷ Perhaps it was rather symptomatic that the review of Kuhn’s *Structure* never appeared in *Minerva*: Kuhn’s work was largely irrelevant to the “*Minerva* debate.”

In the Soviet Union, Kuhn’s *Structure* was also read with an eye to the debates on the social and political implications of the growth of science and its role in society in the atomic and post-atomic age. The STR theory provided a specific context for reading Kuhn in the Soviet Union. During the 1960s, Kuhn’s *Structure* didn’t evoke any particular interest in the Soviet Union, although it was discussed and sympathetically reviewed almost immediately after its publication. In 1963, Henry Guerlac, a Cornell University historian of science, visited Leningrad’s branch of the Institute for the History of Science and Technology and gave a lecture on “the development of the history of science in the USA.”¹⁰⁸ In 1965, the IHST researcher Ludmila Markova published first Soviet review of *Structure*, offering a sympathetic summary. Markova emphasized *Structure*’s significance as a turning point for the history of science and noted its proximity to Marxist thought, characterizing Kuhn’s book as a “dialectical interpretation” of the revolutions in science.¹⁰⁹ However, despite these interventions, it was not until the mid 1970s that Soviet reaction to Kuhn reached momentum with the publication of a Russian translation in 1975. The translation triggered a broad discussion of Kuhn’s book among philosophers and scientists.

In many respects the Soviet discussion of Kuhn’s work in the 1970s resembled the Western one.¹¹⁰ Soviet philosophers and scientists were troubled, just as their Western

counterparts were, by the "incommensurability" thesis. They argued that scientific change was a rational and logical choice, not a somewhat mysterious and irrational "gestalt switch." Kuhn's failure to discuss the sources of new knowledge was also criticized. From the perspective of Soviet critics, Kuhn's concept plausibly accounted for the survival of paradigms but didn't provide a sufficient explanation for the arrival of new paradigms.

Many Soviet reviewers pointed out that Kuhn's model of scientific change wasn't new—that Marx, Engels, and Lenin had recognized early on that the development of science occurs through periodic radical transformations, rather than through gradual grafting of new knowledge onto old. Most explicitly, critics pointed out, the view that the development of science occurs through leaps can be found in Engels' "law" of the transformation of quantitative changes into qualitative changes. Kedrov put it bluntly: "Kuhn put forth a view, long established in Marxism, that progress in science occurs through periodic shifts from the evolutionary to the revolutionary periods of development [of science], through the constant transition from one step to the next one, in the infinite progressive path to the absolute truth."¹¹¹ At the same time, Soviet critics noted that the apparent proximity between Kuhn's concept of "paradigm shifts" and the laws of dialectics was deceptive, since Kuhn's concept was largely concerned with how scientists achieve *agreement* as to what is true, rather than with how science *produces* truth—the objective knowledge of reality. One of Kuhn's critics put it this way:

It is not difficult to find certain points where Kuhn's concept comes into contact with dialectical materialist theory of knowledge. These points of apparent proximity between the two include the implied interconnection and interdependence of theoretical and experimental practices in science, the protest against the absolutization of logical methods of studies of science, the assertion of the social conditioning of scientific research, etc. However, it would be a mistake to talk about any proximity between Kuhn's views and the basic tenets of the Marxist theory of knowledge. One of these major tenets lies in the answer to question about the relation between science and truth. This question is out of the scope of Kuhn's analysis, as the notion of "truth" does not play any role in his concept.¹¹²

The most prominent criticism, however, concerned Kuhn's focus on the universal features (structures) of scientific revolutions. As many Soviet critics emphasized, Kuhn's analysis, seeking for universal patterns in the development of science, didn't account for the unique features of individual revolutions. As the physicist Vitalii Ginzburg, put it, "Kuhn's scientific revolutions are as alike as two drops of water."¹¹³ However, Ginzburg argued, the refutation of phlogiston theory is *qualitatively* different from the challenges to Newtonian mechanics by the theory of relativity, and the conflation of these two different phenomena leads to relativism, to which Ginzburg strongly objected.

Ginzburg's 1976 article opened a discussion of Kuhn's *Structure* on the pages of *Priroda*, the Soviet Union's leading popular-science magazine. The concluding article in this discussion was written by Kedrov, by this time the most prolific Soviet writer on scientific revolutions. Kedrov agreed with Ginzburg that "each scientific revolution must be studied separately as a unique and non-repetitive phenomenon."¹¹⁴ Not only do revolutions differ across time, Kedrov reasoned; there are also significant differences between revolutions in different scientific disciplines, each having its own character, subject matter, and relation to other disciplines. As Kedrov pointed out, the problem with Kuhn's incommensurability thesis derived from his failure to distinguish qualitative differences between revolutions.

By this time, Kedrov had developed a comprehensive "typology" of revolutions in science in his several book-length accounts, arguing that revolutions in different centuries have different causes, effects and outcomes. One of his books was characteristically titled *Scientific Revolution: Substance. Typology. Structure. Mechanism. Criteria*.¹¹⁵ Kedrov distinguished and characterized four "types" of scientific revolutions. The first "type" was the Copernican revolution characterized by Kuhn. Then there was the "Kantian Revolution" that forged the ideas of evolution. Late in the nineteenth century and early in the twentieth, the "New Revolution in the natural sciences" consolidated representations of nature based on mathematical abstractions and probability. Finally, there was the Scientific-Technological Revolution, a new phenomenon that could not be understood by reducing its causes, effects, and outcomes to the previous revolutions in science. Thus, Kedrov contended, "Although we could say, following Kuhn, that in each case there was a radical break with the system of interconnected concepts and views (what Kuhn calls "paradigms"), such a generalized (and hence abstract) approach can hardly be fruitful for the study of the revolutionary development of science."¹¹⁶

Overall, Kuhn's work was received in the Soviet Union with sympathetic interest but without any particular enthusiasm. One of the reasons was the existence of a competing discourse of scientific revolutions—the STR. The STR theorists, many of whom participated in promoting and institutionalizing the new field of *naukovedenie*—a Soviet version of "science studies"—were interested almost exclusively in the period since World War II. For them, history was happening here and now, in the post-atomic age, and promised a revolutionary transformation of the world. Politically primitive and counterrevolutionary, Kuhn, with his focus on normal science as a stabilizing social practice, was largely irrelevant for the theorists of the STR.

Conclusion

With the changes in the political economy of science that occurred during and after World War II—changes associated with science's dramatically increased economic dependence on public resources and the military patronage—the conventions for representing scientific enterprise had also changed on both sides of the Iron Curtain. By the early 1960s, scientists and social analysts had responded to the political challenges of the time by insisting on the image of science as a concrete, historical, and interacting community of scientist-citizens rather than timeless and universal "science" without national or political affiliations. The historian David Hollinger characterized the period as "a watershed in the history of discourse about science," which moved away from the traditional focus of philosophy of science on science method and intellectual history as foremost preoccupation of history of science and toward a focus on science's social, political, and economic contexts and aspects.¹¹⁷

The discussions of the social and political consequences of Big Science had contributed to this "paradigm shift" in the perception of science. In the United States, the loosely connected network of intellectuals associated with the Congress of Cultural Freedom contributed to a construction of a public space in which the relations between science and politics were debated, focusing the discussion on organization of science, science policy, and the planning of science. With the development of Science Studies as an academic discipline, the issues of science policy became marginalized, with science studies primarily focused on knowledge production—the central concern of Science Studies since the 1970s.¹¹⁸

The establishment of the journal *Science Studies* in 1970 marked the end of this earlier version of postwar "science studies"—"Big Science studies." Perhaps the founders of *Science Studies* didn't fully realize the extent to which the launch of this journal was a slap in the face to the CCF-associated "Big Science studiers." The founders of *Science Studies* (later renamed *Social Studies of Science*), Roy MacLeod and David Edge, both recorded the moment of the creation of the new journal in their recollections. As Edge recollected, John Maddox—then the head of Macmillan's journals division—had been enthusiastic about the idea of a new quarterly journal called *Science Studies*, even hoping that it would become a weekly. Edward Shils' reaction was furious, however. He "wanted to sabotage us, [saying] that we were committing academic suicide. ... He wanted to stop us. ... He did not stop us, but he kept treating us like we were ... I don't know ... just nuts."¹¹⁹ MacLeod presented a more polite version of the same story in his published memoir: "Edward Shils cordially discouraged me from doing anything of the sort. There was simply not enough good material, he said, with the implication that anything 'good' he would publish in *Minerva* himself."¹²⁰ Shils gave his own account of this moment in a letter to Shepard Stone in 1970, immediately after the launch of *Science Studies*: "He [John Maddox] undertakes

to establish another periodical on more or less the same subject as *Minerva* ... and he had one of his handy-girls prepare a promotional letter for this journal (it is called, I think, *Science Studies*) in which it is alleged that there is no competition between the magazines because *Minerva* deals only with relations between science and government. This is, of course, a caricature of the wide range of subjects treated by *Minerva*."¹²¹

In hindsight, the vision of "science studies" that Shils and his associates were advocating differed from academically linked Science Studies programs established at the universities in the United States and the United Kingdom in the 1960s and the 1970s. It was different, too, from "science studies" or *naukovedenie* in the Soviet Union. Yet, I would argue, these alternative approaches to studies of science, in which scientific achievements and growth appeared inseparable from issues of science policy, organization of science, science governance, ethics of science, and the planning of science, constituted an important part of the "pre-history" of Science Studies as we know it today, by establishing public forums for analysis and debate.

The case of the CCF and the "*Minerva* debate" can also illustrate the complexities and ambiguities of "cultural cold wars." The outcomes of discussions on science initiated under the auspices of the CCF were *shaped* by the organization's explicit political agenda but were not directly, let alone unequivocally, *determined* by the political demands. Although the studies of science during the Cold War encapsulated the political concerns and anxieties of the time, there was no single Cold War "party line." The CCF and its engagement in "science studies" presents a story akin to other social sciences that have received ample support during the Cold War: Area Studies, behavioral science, human relations, development studies, American Studies, and a host of other "studies" and interdisciplinary "clusters" that served national interest but not necessarily in predictable ways.¹²² The CCF intellectuals' claims of being "independent" and "free" in their thinking would not survive the 1960s. The revelation of the CIA's sponsorship of the CCF shattered comfortable assumptions of scholars in the service of the state. Yet their quest for "middle ground," reconciliation, and compromise, as part and parcel of their conceptions of scholarship and service to the state, was effectively shifting the debate away from simplistic Cold War narratives of East-West competition.

As the term "Big Science" gained currency after the end of the Cold War, it gradually lost its political, connotations. This chapter, by exploring the context in which the concept of "Big Science" was framed originally—in Cold War America, as well as in the post-Stalinist Soviet Union—suggests that extending the notion of "Big Science" to later periods may be problematic. Although specific "characteristics" of Big Science as a mode of organization of science can be discerned as a set of traits applicable to different examples of large-scale scientific enterprises in the past, such extension draws attention away from the explicit Cold War connotations of this notion, and from what

Stephen Toulmin called the "political instrumentality" of Big Science shaped by the political economy of the Cold War.

Notes

1. See Peter Galison, "The Many Faces of Big Science," in *Big Science: The Growth of Large-Scale Research*, ed. Peter Galison and Bruce Hevly (Stanford University Press, 1992); James H. Capsheew and Karen Rader, "Big Science: Price to the Present," *Osiris* 7 (1992): 3–25; Steven Shapin, *Scientific Life* (University of Chicago Press, 2008); Jon Agar, *Science in the Twentieth Century and Beyond* (Polity, 2012).
2. Agar, *Science in the Twentieth Century*, 330 (emphasis added). This distinction is also made in Capsheew and Rader's "Big Science."
3. Edward Shils, "Minerva [editorial]," *Minerva* 1, no. 1 (1962): 5–17, at 16.
4. *Ibid.*, 9.
5. *Ibid.*, 12.
6. Alvin M. Weinberg, "Impact of Large-Scale Science on the United States," *Science* 134 (1961): 161–164; Derek de Solla Price, *Science Since Babylon* (Yale University Press, 1961).
7. Weinberg, "Impact of Large-Scale Science," 162. On Weinberg's championing of the view of Big Science as a pathological condition and his use of the metaphors of "disease," see Capsheew and Rader, "Big Science."
8. On scientists' reservations and criticism of Big Science, see Shapin, *Scientific Life*, 80–89. On biologists' reactions on different modes of Big Science in the 1960s, see Elena Aronova, Karen Baker, and Naomi Oreskes, "Big Science and Big Data in Biology: From the International Geophysical Year through the International Biological Program to the Long Term Ecological Research (LTER) Network, 1957–Present," *Historical Studies in the Natural Sciences* 40, no. 2 (2010): 183–224.
9. Weinberg, "Impact of Large-Scale Science," 161.
10. *Ibid.*
11. Edward Shils to Alvin Weinberg, November 2, 1962, Minerva Records, Special Collections Research Center, Regenstein Library, University of Chicago (hereafter cited as Minerva Records), box 1, folder 12.
12. Alvin Weinberg to Edward Shils, November 15, 1962, Minerva Records, box 1, folder 12.
13. Edward Shils to Alvin Weinberg, November 28, 1962, Minerva Records, box 1, folder 12.
14. Alvin Weinberg, "Criteria for Scientific Choice," *Minerva* 1/2 (1963): 159–171.
15. *Ibid.*, 159.
16. *Ibid.*, 171.

17. Ibid., 170.
18. Polanyi was the source of Shils' perhaps deepest intellectual inspiration, which, as Stephen Turner put it, "cannot be reduced to a model of 'influence.' ... [Rather,] it amounted in a way to a kind of dialectical partnership that stimulated Shils' thought and Polanyi's as well." Stephen P. Turner, "Obituary for Edward Shils," *Tradition and Discovery* 22, no. 2 (1996): 5–10. See also Louis H. Swartz, "Michael Polanyi and the Sociology of a Free Society," *The American Sociologist* 29 (1998): 59–70.
19. Michael Polanyi to Edward Shils, 15 November 1962, Minerva Records, box 1, folder 4. As Shils later admitted, Polanyi became less interested in the journal when it became more focused on "science-policy." Edward Shils, "A Great Citizen of the Republic of Science: Michael Polanyi, 1892–1976," *Minerva* 14, no. 4 (1976): 1–5.
20. Michael Polanyi, "The Republic of Science: Its Political and Economic Theory," *Minerva* 1, no. 1 (1962): 54–73.
21. Ibid. For a thorough discussion of Polanyi's political philosophy, see Mary Jo Nye, *Michael Polanyi and His Generation: Origins of the Social Construction of Science* (University of Chicago Press, 2011).
22. Polanyi, "The Republic of Science," 54.
23. As Mary Jo Nye had demonstrated, these views were rooted in Polanyi's career and the experiences he had in the 1920s and the early 1930s as a physical chemist in the Kaiser-Wilhelm-Gesellschaft Institutes in Berlin-Dahlem—the institutions that enjoyed the government support at the same time not limiting the scientists' autonomy. Mary Jo Nye, "Historical Sources of Science-as-Practice: Michael Polanyi's Berlin," *Historical Studies in the Physical and Biological Sciences* 37, no. 2 (2007): 409–434.
24. Stephen Toulmin, "The Complexity of Scientific Choice," *Minerva* 2 (1964): 343–359. For a more detailed discussion of the "Minerva debate," see Elena Aronova, "The Congress for Cultural Freedom, Minerva, and the Quest for Instituting 'Science Studies' in the Age of Cold War," *Minerva* 50 (2012): 307–337.
25. Toulmin, "The Complexity of Scientific Choice," 354.
26. Michael Polanyi, "The Growth of Science in Society," *Minerva* 5, no. 4 (1967): 533–545, at 543–544.
27. Alvin Weinberg, "Scientific Choice and the Scientific Muckrakers: Review Article," *Minerva* 7, no. 1–2 (1968): 52–63, at 61.
28. Alvin Weinberg, "Origins of Criteria for Scientific Choice," *Current Contents* 32 (1991): 18. For a discussion, see Aant Elzinga, "The Rise and Demise of the International Council for Science Policy Studies (ICSPS) as a Cold War Bridging Organization," *Minerva* 50 (2012): 277–305, at 296.
29. Weinberg, "Criteria for Scientific Choice," 160.
30. Ibid.

31. Alvin Weinberg, "Edward Shils and the 'Governmentalisation' of Science," *Minerva* 34 (1996): 39–43, at 42.
32. On the history of the CCF, see Frances Stonor Saunders, *The Cultural Cold War: The CIA and the World of Arts and Letters* (New Press, 2000); Volker R. Berghahn, *America and the Intellectual Cold Wars in Europe: Shepard Stone between Philanthropy, Academy and Diplomacy* (Princeton University Press, 2001); Giles Scott-Smith, *The Politics of Apolitical Culture: The Congress for Cultural Freedom, the CIA and Post-War American Hegemony* (Routledge, 2002); Peter Coleman, *The Liberal Conspiracy* (Free Press, 1989).
33. Cited in Coleman, *The Liberal Conspiracy*, 56.
34. On Koestler's "hard-line" in the early days of the CCF, see Berghahn, *America and the Intellectual Cold Wars*; Scott-Smith, *The Politics of Apolitical Culture*.
35. Michael Polanyi to V. B. Karnik, January 11, 1961, Michael Polanyi Papers, Special Collections Research Center, Regenstein Library, University of Chicago (hereafter cited as Polanyi Papers), b. 6, f. 1.
36. Scott-Smith, *The Politics of Apolitical Culture*, 139.
37. Michael Polanyi, "CCF, Memo," 19 November 1961, Polanyi Papers, b. 6, f. 6.
38. K. A. Jelenski, "Introduction," in *History and Hope: Progress and Freedom; The Berlin Conference of 1960*, ed. Jelenski (Routledge & Kegan Paul, 1962), 1.
39. Memo "Study Group of the Committee for Science and Freedom held in Paris 29 August to 1 September 1956," Polanyi Papers, b. 33, f. 10.
40. Rebecca Lemov, "'Hypothetical Machines': The Science Fiction Dreams of Cold War Social Science," *Isis* 101 (2010): 401–411.
41. "Memorandum" (n.d.), International Association for Cultural Freedom Papers, Special Collections Research Center, Regenstein Library, University of Chicago (hereafter cited as IACF Papers), Series III, box 99, folder 1.
42. *Ibid.*
43. *Ibid.*
44. "CCF Study Groups," Series III, bb. 27, 28, 29. IACF Papers.
45. "Memorandum," (n/d), IACF Papers, Series III, b. 99, f. 1.
46. "Committee on Science and Freedom. Report on the First Year's Activities: July 1954–August 1955." Committee on Science and Freedom Study Group, Agenda (Paris, 1956). IACF Papers, Series III, Box 12, folder 4. On Polanyi's Society for Freedom in Science, see Nye, *Michael Polanyi and His Generation*.
47. "Hamburg Congress on Science and Freedom, General Summary," IACF Papers, box 5, folder 7.

48. "Shils Comments on Polanyi," Science and Freedom Congress, Hamburg 1953, IACF Papers, box 5, folder 2.
49. Committee on Science and Freedom Study Group, Agenda (Paris, 1956). IACF Papers, Series III, b. 12, f. 4.
50. Edward Shils, "Outline of the General Conference of the Congress for Cultural Freedom," April 1959, International Association for Cultural Freedom/Congress for Cultural Freedom/International Council on the Future of the University Records, Special Collections Research Center, Regenstein Library, University of Chicago (hereafter cited as IACF/ICFU Papers), b. 1.
51. Ibid.
52. CCF Study Groups, IACF Papers, Series III, b. 27–29.
53. Edward Shils to A. K. Brohi, 11 April 1960, IACF/ICFU Papers, b. 1.
54. Edward Shils to J. Robert Oppenheimer, 12 July 1960, IACF/ICFU Papers, b. 1.
55. Cited in Coleman, *The Liberal Conspiracy*, 9.
56. On the CIA and its involvement in funding scientific research in various fields, see Ronald Doel and Allan A. Needell, "Science, Scientists, and the CIA: Balancing International Ideals, National Needs, and Professional Opportunities," in *Eternal Vigilance? 50 Years of the CIA*, ed. Rhodri Jeffreys-Jones and Christopher Andrew (Frank Cass, 1997). As Doel and Needell show, scientists were generally willing to aid the CIA in providing scientific intelligence on atomic, biological, and chemical weapons as well as basic science fields in the 1950s. Discussing the ways different scientists involved in intelligence gathering had tried to negotiate, reconcile and/or in various ways struggled with fundamental differences between the "ideals and values" of intelligence gathering and those of science, Doel and Needell argued that these experiences "have profoundly altered the image and practice of science in postwar America."
57. Cited in Christopher Lasch, *The Agony of the American Left* (Andre Deutsch, 1970).
58. Lasch, *The Agony of the American Left*, 104–105.
59. Sidney Hook, *Out of Step: An Unquiet Life in the 20th Century* (Harper & Row, 1987), 451.
60. "Copy of the text of the letter sent on May 4th to the Editor of the *New York Times* by John Kenneth Galbraith, George Kennan, Arthur Schlesinger, Jr." [1967], IACF/ICFU Papers, b. 1.
61. Edward Shils to Crawford, Goodwin, n/d. IACF/ICFU Papers, b. 1, f. 12.
62. Ibid. As a countermeasure against the accusations thrown by the press, Shils suggested a project aimed at producing a well-documented history of the CCF, which would imply the organization of the archive of the CCF records documenting its activities and the oral history interviews with the "intellectual figures who played leading roles in the history of the Congress," such as Michael Polanyi, Raymond Aron, Arthur Koestler, Willy Brandt, George Kennan, Daniel Bell, as well as with the members of the CCF executive staff: Michael Josselson, Nicolas Nabokov,

Francois Bondy, Konstantin (Cot) Jelenski, Pierre Emmanuel, and Melvin Lasky. IACF/ICFU Papers, b. 1, f. 12.

63. Michael Polanyi to Pierre Emmanuel, 9 April 1968, Polanyi Papers, b. 6, f. 13.

64. Michael Polanyi to Raymond Aron, May 9, 1967, Polanyi Papers, b. 6, f. 10.

65. Michael Polanyi to Pierre Emmanuel, 9 April 1968, Polanyi Papers, b. 6, f. 13.

66. "Press Release, Monday, October 2, 1967," Polanyi Papers, b. 6, f. 10.

67. On the CCF after its reorganization in 1967, see Aronova, "The Congress for Cultural Freedom."

68. Howard Brick, "Optimism of the Mind: Imagining Postindustrial Society in the 1960s and 1970s," *American Quarterly* 44, no. 3 (1992): 348–380.

69. Transcript of Proceedings of the conference "Post-Industrial Society and Cultural Diversity," June 12–14, 1970. IACF Papers, Series III, b. 80, f. 3.

70. Pyotr Kapitsa, "Slovo o progresse," *Pravda*, January 20 1966; Kapitsa, "Uchenyi i plan," *Literaturnaya Gazeta*, March 5, 1966; Kapitsa, "Scientific Policy in the USSR: The Scientist and the Plans," *Minerva* 4, no. 4 (1966): 555–560.

71. Edward Shils, "Kapitsa's views on how science should be planned," *Minerva* Records, box 8, folder 10.

72. Slava Gerovitch, *From Newspeak to Cyberspeak: A History of Soviet Cybernetics* (MIT Press, 2002).

73. Kapitsa, "Scientific Policy in the USSR," 556.

74. *Ibid.*, 559.

75. Edward Shils to Marion Bieber, n/d [1966], *Minerva* Records, box 7, folder 14.

76. Michael Polanyi, opening address, 1953 Congress "Science and Freedom," Hamburg, IACF Papers, box 5, folder 7.

77. For an overview of the Soviet economy, see Philip Hanson, *The Rise and Fall of the Soviet Economy and Economic History of the USSR from 1945* (Longman, 2003).

78. Although the literal translation of "tekhnicheskaya" should be "technical" (in contrast to "tekhnologicheskaya" that stands for "technological" in Russian) we are using a standard rendering in English literature on Scientific-Technological Revolution.

This section is based on the previously published article: Elena Aronova, "The Politics and Contexts of Soviet Science Studies (*Naukovedenie*): Soviet Philosophy of Science at the Crossroads," *Studies in East European Thought* 63, no. 3 (2011): 175–202.

79. In the 1970s and the 1980s the literature on the STR was thoroughly reviewed by American Sovietologists and political analysts. See Erik P. Hoffmann and Robbin F. Laird, "The

Scientific-Technological Revolution" and *Soviet Foreign Policy* (Pergamon, 1982); Hoffmann and Laird, *The Politics of Economic Modernization in the Soviet Union* (Cornell University Press, 1982); Hoffmann and Laird, *Technocratic Socialism: The Soviet Union in the Advanced Industrial Era* (Duke University Press, 1985); Cyril Black, "The Scientific-Technological Revolution: Economic to Scientific Determinism?" (occasional paper presented at Kennan Institute for Advanced Russian Studies, 1979); Arnold Buchholz, "The Role of the Scientific-Technological Revolution in Marxism-Leninism," *Studies in Soviet Thought* 20 (1979): 145–164; Arnold Buchholz, "The Scientific-Technological Revolution (STR) and Soviet Ideology," *Studies in Soviet Thought* 30 (1985): 337–346; Erik P. Hoffmann, "Soviet views of 'Scientific-Technological Revolution,'" *World Politics* 30 (July 1978): 615–644; Friedrich Rapp, "Soviet-Marxist Philosophy of Technology," *Studies in Soviet Thought* 29 (1985): 139–150.

80. The title of the volume is reminiscent of *Science at the Crossroads*, Papers from the Second International Congress of the History of Science and Technology, ed. Nikolai I. Bukharin et al. (Frank Cass, 1971; originally published in 1931), of which Boris Hessen's paper on Newton is the most famous legacy. (I thank an anonymous reviewer for pointing out this connection.) Although *Civilization at the Crossroads* didn't make explicit references to that volume, it appears to be an homage to Bukharin, since Richta's conception contained the theoretical redefinitions of Marxist doctrine along the lines of Bukharin's theoretical position on the role of productive forces and the function of science (both natural and social sciences) in society. On Richta's theoretical views, see Vitezslav Sommer, "From 'Active Superstructure' to Culture of Socialist Experts: Czechoslovak Social Sciences and the Construction of 'Scientific' in the 1950s and 1960s," paper presented at workshop on Politics and Contexts of Science Studies during the Cold War and Beyond, Greifswald, 2012. On Bukharin's views, see Helena Sheehan, *Marxism and the Philosophy of Science* (Humanities Press, 1985).

81. Radovan Richta et al., *Civilization at the Crossroads: Social and Human Implications of the Scientific and Technological Revolution* (Australian Left Review Publications, 1969), 19.

82. Vladimir Kusin, *The Intellectual Origins of the Prague Spring* (Cambridge University Press, 1971).

83. Hanson, *The Rise and Fall of the Soviet Economy*, 92.

84. Roy Boyd and Tony Caporale, "Scarcity, Resource Price Uncertainty and Economic Growth," *Land Economics* 72, no. 3 (1992): 326–335; anonymous, "International Issues: Henry Kissinger's Views on Possible US Military Intervention Against the Oil Producers," *Time* 105 (January 20, 1975): 34–35; Antony E. Reinsh, Igor Lavrovsky, and Jennifer I. Considine, *Oil in the Former Soviet Union: Historical Perspective Long-term outlook* (Canadian Energy Research Institute, 1992).

85. On the history of the Fiat-Togliatti plant, see Lewis H. Siegelbaum, *Cars for Comrades: The Life of the Soviet Automobile* (Cornell University Press, 2008).

86. Ibid. In the American debates over this foreign policy development, the gains made available to the Soviet Union from easier access to Western machinery and know-how became the center of controversies over détente and Nixon-Kissinger strategy. See "International Development, Trade Policies, 1969–1972," <http://www.state.gov/r/pa/ho/frus/nixon/iv/15710.htm>.

87. Genrikh N. Volkov, *Man and the Challenge of Technology* (Novosti Press Agency Publishing House, 1972), 37.

88. Bonifatii M. Kedrov, Semen R. Mukulinsky, and Ilia T. Frolov, *Nauchno-technicheskaia revoliutsiia i sotsial'nyi progress* (Politizdat, 1972).

89. For the detailed account, see Aronova, "The Politics and Contexts of Soviet Science Studies." On the institutionalization of history of science in the Soviet Union in the 1920s and the 1930s, see M. S. Bastarkova, "Iz istorii razvitiia istoriko-nauchnykh issledovanii," *Voprosy Istorii Estestvoznaniia i Tekhniki* 61–63 (1978): 34–47; Semen S. Ilizarov, *Institut Istorii Estestvoznaniia i Tekhniki im. S. I. Vavilova, 1953–1993* (Nauka, 1993); Aleksandr N. Dmitriev, "Institut Istorii Nauki i Tekhniki v 1932–1936 gg," *Voprosy Istorii Estestvoznaniia i Tekhniki* 1 (2002): 3–36; Natalia L. Gindilis, "Predystoriia otechestvennogo naukovedeniia," *Voprosy Istorii Estestvoznaniia i Tekhniki* 2 (2009): 160–178.

90. As Slava Gerovich has argued in his account of the strategies of Soviet historians of science in dealing with the Cold War, Soviet historiography of science followed the political and social evolution of Soviet society. See Gerovitch, "Writing History in the Present Tense: Cold War-Era Discursive Strategies of Soviet Historians of Science and Technology," in *Universities and Empire: Money and Politics in the Social Sciences during the Cold War*, ed. Christopher Simpson (New Press, 1998).

91. Kedrov, an important figure in Soviet history and philosophy of science, had been writing on a range of issues and had been involved in major Soviet political debates over science since the 1920s; for the detailed discussion of his work and career, see Aronova, "The Politics and Contexts of Soviet Science Studies." Kedrov's most original work in the history of science was his meticulous hour-to-hour reconstruction of Mendeleev's discovery of the periodic law. (Kedrov called it "a microanatomy of scientific discovery.") As Michael Gordin pointed out, in that work Kedrov emphasized the role of textbooks in the construction of the periodic system—a line of reasoning quite resonant with the present day science studies approaches: see Michael Gordin, "The organic roots of Mendeleev's periodic law," *Historical Studies in the Physical Sciences* 32, no. 1 (2002): 263–290.

92. Many historians and literary critics retreated to a descriptive and factological style, or to "textology" (the publication of original texts with only minimal commentary and with no interpretation). For a discussion of Pushkin scholars during Pushkin Centennials in the late 1930s, when Pushkin was redefined and mythologized as a Soviet hero, see Karen Petrone, *Life Has Become More Joyous, Comrades: Celebrations in the Time of Stalin* (Indiana University Press, 2000), 113–149.

93. Prilozhenie "O napravlenii nauchnykh issledovanii i strukture Instituta istorii estestvoznaniia i tekhniki AN SSSR" k postanovleniiu Presidiuma AN SSSR ot 12 oktiabria 1962 (Archive of the Institute for the History of Science and Technology, Moscow, Russia, hereafter cited as IHST Papers).

94. Prilozhenie "O napravlenii nauchnykh issledovanii i strukture Instituta istorii estestvoznaniia i tekhniki AN SSSR" k postanovleniiu Presidiuma AN SSSR ot 12 oktiabria 1962 (IHST Papers).

95. The “collective volumes” on the STR published by IHST researchers or under the supervision of IHST included the following: A.A. Kuzin, N.N. Stokova, S.V. Shukhardin et al., *Sovremennaia nauchno-technicheskaia revoliutsiia. Istoricheskoe issledovanie* (Nauka, 1967); *Sovremennaia nauchno-technicheskaia revoliutsiia* (1967); *Nauchno-technicheskaia revoliutsiia i sotsial'nyi progress* (1972); *Chelovek—Nauka—Tekhnika (opyt marksistskogo analiza nauchno-technicheskoi revoliutsii)* (1973); *Nauchno-technicheskaia revoliutsiia i izmenenie struktury nauchnykh kadrov SSSR* (1973); *Nauchno-technicheskaia revoliutsiia i obshchestvo* (1973); *Partiia i sovremennaia nauchno-technicheskaia revoliutsiia v SSSR* (1974). The names of only a few authors, usually the supervisors, were listed on the title pages, and some had no authors' names at all. The 1973 volume *Chelovek—Nauka—Tekhnika* was the collaboration with Czech philosophers who were involved in the reform movement and were displaced or disgraced after the crushing of the Prague Spring.
96. Despite the suppression of the Prague reform movement, the Soviet and Czechoslovak academies continued to collaborate, but the volume was published without any names listed as the main authors, and was presented as the “combined product of a collective body of authors, members of three institutes.” See *Chelovek—Nauka—Tekhnika (opyt marksistskogo analiza nauchno-technicheskoi revoliutsii)* (Politizdat, 1973), 3.
97. Hoffmann and Laird, *The Politics of Economic Modernization*.
98. Black, “The Scientific-Technological Revolution.”
99. Similar discussions were taking place in both West and East European countries. On Big Science in Germany, see Gerhard A. Ritter, *Großforschung und Staat in Deutschland. Ein historischer Überblick* (Beck, 1992); Margit Szöllösi-Janze, *Geschichte der Arbeitsgemeinschaft der Grossforschungseinrichtungen: 1958–1980* (Campus, 1990).
100. Svetlana I. Ikonnikova, *Teoriia “postindustrial'nogo obshchestva”: budushchee chelovechestva I ego burguaznye tolkovateli* (Mysl', 1975).
101. Buchholz, “The role of the scientific-technological revolution.”
102. Georgii Kh. Shakhnazarov, *Fiasco futurologii* (Politizdat, 1979), 44–45.
103. Aronova, “The Politics and Contexts of Soviet Science Studies.”
104. David Hollinger, “Science as a Weapon in *Kulterkämpfe* in the United States During and After World War II,” *Isis* 86, no. 3 (1995): 440–454, at 453.
105. Ibid.
106. David Hollinger, *Science, Jews, and Secular Culture: Studies in Mid-Twentieth-Century American Intellectual History* (Princeton University Press, 1996), 101.
107. Edward Shils to Stephen Toulmin, February 19, 1964, Minerva Records, box 3, folder 10.
108. Otchet o rabote Leningradskogo otdeleniia Instituta istorii estestvoznaniia i tekhniki AN SSSR v 1963 g. (IHST Papers, Protokoly zasedaniy direktcii).

109. Ludmila A. Markova, "Problema revolucii v istorii nauki," *Voprosy Filosofii* 9 (1965): 178–182.
110. On the reception and the responses to Kuhn's *Structure* in the United States, see John H. Zammito, *A Nice Derangement of Epistemes: Post-Positivism in the Study of Science from Quine to Latour* (University of Chicago Press, 2004).
111. Bonifatii M. Kedrov, *Lenin i metodologicheskie voprosy istorii nauki* (Znanie, 1969).
112. V. M. Legostaev, "Filosofskaya interpretatsiya konceptcii nauchnogo razvitiia Tomasa Kuna," *Voprosy Filosofii* 11 (1972): 129–136, at 136.
113. Ginzburg considered Kuhn's book "trivial" in some respects though interesting and largely correct (from the point of view of a physicist). "Kuhn's scientific revolutions," he wrote, "are as alike as two drops of water, and, moreover, practically any change in science, however minor it is, can be called a revolution ... within a particular scientific community which can be as little as 25 individuals. ... If we would take the same approach to social changes, we should call any coup d'état or even a reorganization of an office consisting of 25 staff members, a revolution." Vitalii L. Ginzburg, "Kak razvivaetsia nauka?" *Priroda* 6 (1976): 73–85, at 78.
114. Bonifatii M. Kedrov, "O revoliutsionnom kharaktere razvitiia estestvoznaniia," *Priroda* 10 (1976): 68–71.
115. Bonifatii M. Kedrov, *Nauchnye revoliucii: Sushchnost'. Tipologiya. Struktura. Mekhanizm. Kriterii* (Znanie, 1980). See also Bonifatii M. Kedrov, *Lenin i revoliuciia v estestvoznanii v 20 v. Filosofskie i estestvennye nauki* (Moskva, 1969) and Bonifatii M. Kedrov, *Lenin i nauchnye revoliucii. Estestvennye nauki. Fizika* (Znanie, 1980).
116. Kedrov, "O revoliutsionnom kharaktere razvitiia estestvoznaniia."
117. Hollinger, *Science, Jews, and Secular Culture*, 101.
118. On the relation between Science Studies and Science Policy Studies, see Aant Elzinga, "Changing Policy Agenda in Science and Technology," in *Handbook of Science and Technology Studies*, ed. Sheila Jasanoff, Gerald E. Markle, James C. Petersen, and Trevor J. Pinch (London, 1995); Elzinga, "The Science-Society Contract in Historical Transformation: With Special Reference to 'Epistemic Drift,'" *Social Science Information* 36, no. 3 (1997): 411–445.
119. David Edge Oral History interview, transcript, British Society for the History of Science Oral History Project "The history of science in Britain, 1945–1965," Brotherton Library, University of Leeds, BSHS 10/8/7, on 52.
120. Roy MacLeod, "David Edge. In Memoriam," *Social Studies of Science* 33, no. 2 (2002): 181–183.
121. Edward Shils to Shepard Stone, 6 October 1970, IACF/ICFU Papers, b. 1, f. 16.
122. David C. Engerman, "Social Science in the Cold War," *Isis* 101 (2010): 393–400.

